CSE2023 ASSIGNMENT #4 (WILL NOT BE GRADED)

Solve the following questions from the course book. (Discrete Mathematics and Discrete Mathematics and Its Applications, 7th Ed. by Kenneth Rosen)

1) Prove that $2^n > n^2$ if n is an integer greater than 4.

2) How many positive integers between 51 and 151 inclusive
   a. are divisible by 4?
   b. are divisible by 6?
   c. are not divisible by either 4 or 6?

3) How many bit strings of length 14 contain at least six 1s and at least five 0s?

4) What is the minimum number of students, each of whom comes from one of the 40 states, who must be enrolled in a university to guarantee that there are at least 80 who come from the same state?

5) The Turkish alphabet contains 21 consonants and 8 vowels. How many strings of ten lowercase letters of the Turkish alphabet contain
   a. exactly three vowel?
   b. exactly four consonants?
   c. at least two vowel?
   d. at least three vowel?
   e. at least three consonants?

6) How many bit strings of length 18 contain at least five 1s and at least two 0s?